A Mechatronic Eye Movement Emulator for Benchmarking AR/VR Eye-Tracking Sensors

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Background

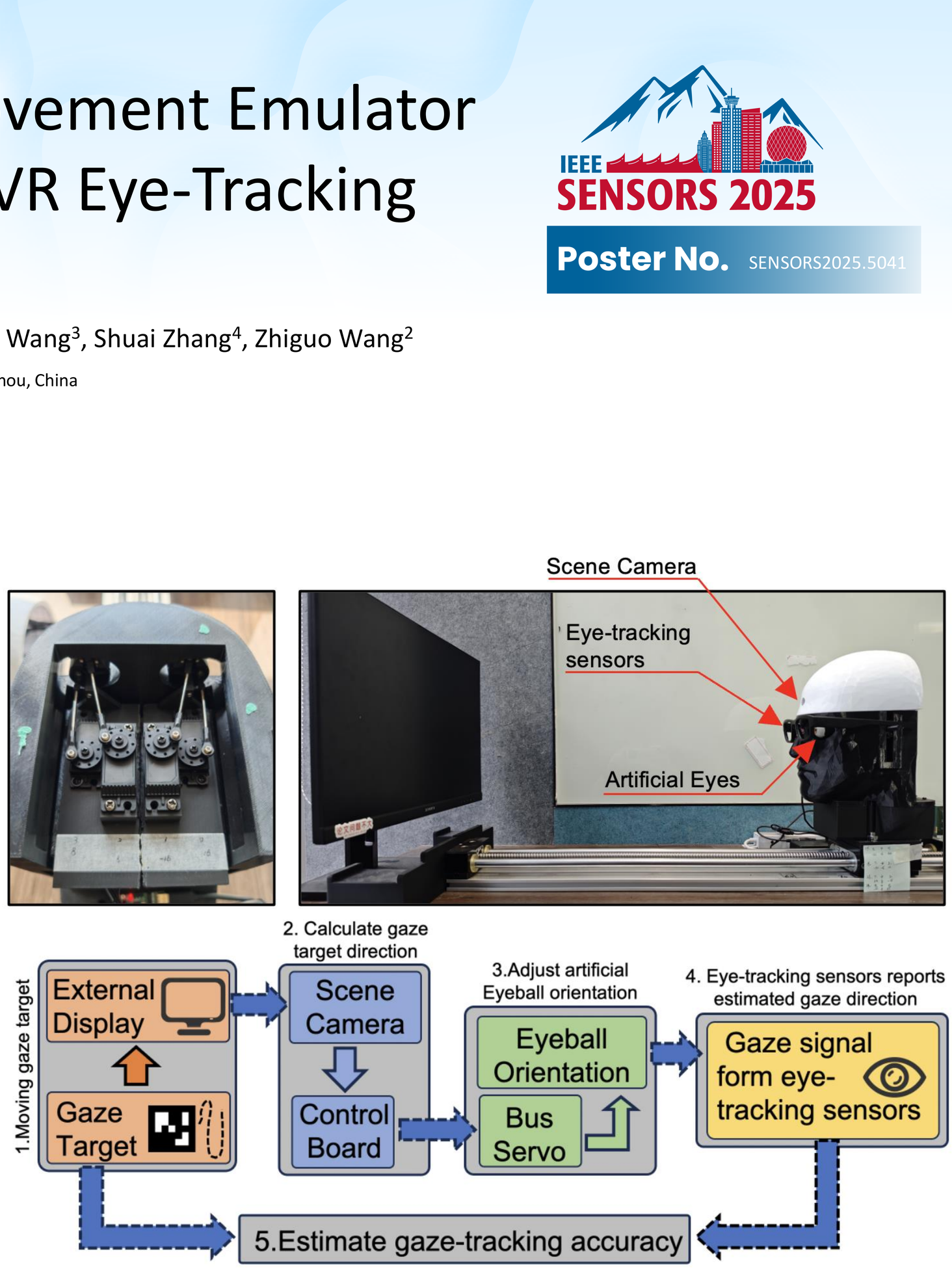
- Eye-tracking sensors are critical components of augmented and virtual reality (AR/VR) systems, underpinning functionalities such as foveated rendering and gaze-based interactions.
- Mainstream evaluation methods for eye-tracking sensors often rely on synthetic eye images, which are confined to camera-based systems.
- The lack of standardized, hardware-agnostic benchmarking tools hampers cross-platform comparisons.
- We present a mechatronic eye movement emulator with a bioinspired design to benchmark diverse sensors.

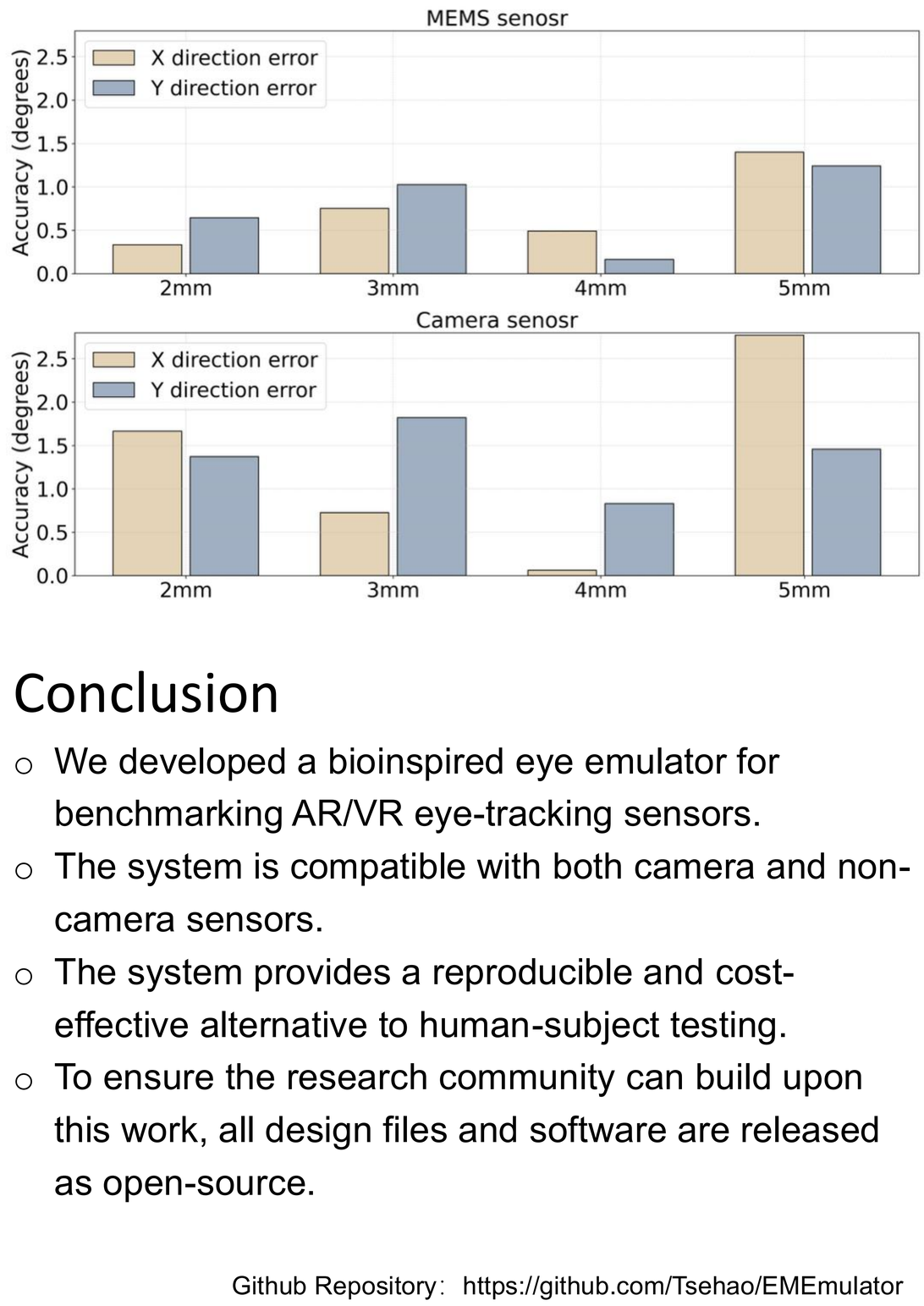
Methods

- System Components
  - A 3D-printed head model.
  - A pair of artificial eyes with a 25 mm diameter, a 12 mm iris, a 2–5 mm pupil, and a cornea.
  - Servo actuators for controlling eye rotations.
  - Scene camera for tracking of a moving ArUco marker.
- Control & Software
  - Servo control with backlash compensation.
  - Python 3.11, OpenCV 4.11, real-time ArUco tracking.
- Testing Setup
  - Devices: Tobii Pro Glasses 2 (camera-based) and AdHawk MindLink (MEMS-based).
  - Conditions: 2–5 mm pupil diameters, 70 cm viewing distance, gaze target moving in a Lissajous pattern.

Results

- Reliability
  - Gaze shifts were reliably detected by both camera- and MEMS-based eye-tracking sensors.
- Tracking Accuracy
  - MEMS-based MindLink → MAE ≈ 0.74° (H), 0.77° (V).
  - Camera-based Tobii G2 → MAE ≈ 1.31° (H), 1.37° (V).
  - Accuracy varied with pupil sizes (2–5 mm).
  - Measured gaze errors fall within typical ranges for wearable eye-tracking glasses (1–4°).





Conclusion

- We developed a bioinspired eye emulator for benchmarking AR/VR eye-tracking sensors.
- The system is compatible with both camera and non-camera sensors.
- The system provides a reproducible and cost-effective alternative to human-subject testing.
- To ensure the research community can build upon this work, all design files and software are released as open-source.

Github Repository: https://github.com/Tsehao/EMEmulator